

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q18-1032

Eklutna 138 kV
March 02, 2020

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q18-1032. The Interconnection Customer proposes a 151.9 MW natural gas combined cycle facility interconnecting at the Eklutna 138 kV substation in the Pennsylvania area. The request was received on July 22, 2018. The study evaluated system intact and single contingency conditions using the regional planning case, including 334 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 4 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$48,629,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

The Norgate to Vera 138 kV L is overloaded to 127.6% of its 183 MVA emergency rating for the loss of the Caldwell to Carbon Ctr 345 kV Line. The project contributes 7.8% DFAX (11.9 MW) to this overload.

The Eklutna to Elsinore 138 kV is overloaded to 111.1% of its 250 MVA emergency rating for the loss of the Blackrock to Eklutna 345 kV. The project contributes 35.2% DFAX (53.5 MW) to this overload.

Carbon Center 500-345 kV Transformer Bank: 101.3% loading (1,716 MVA) following the outage of the Blackrock Substation to Shawville Sta 345 kV; DFAX = 16.9%.

For the loss of Ft Gibson to Armstrong 500 kV Line, the New Melones to Armstrong 345 kV loads to 100.3% of its 1,061 MVA rating. Project DFAX: 6.8%.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Vera to Norgate 138 kV	Rebuild line with higher-capacity conductor	\$33,951,000	100% (\$33,951,000)
Eklutna to Elsinore 138 kV	Rebuild line with higher-capacity conductor	\$12,926,000	100% (\$12,926,000)
Carbon Center 500-345 kV Transformer Bank	Replace transformer with higher-rated unit	\$23,752,000	6% (\$1,344,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
New Melones to Armstrong 345 kV line	Reconductor line	\$20,739,000	2% (\$408,000)
	Total allocated to project		\$48,629,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.